

SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

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ABSTRACT. We prove that seven open unit equilateral triangles cannot cover a regular unit hexagon. The center and vertices distinguish the covering triangles. Boundary gaps and midpoint coverage reduce the proof to trace-length, area-loss, and finite-enclosure obstructions. The enclosure arguments use six points for a selected boundary gap, four points for a supported midpoint, and nine points when the vertex triangles cover the boundary. A neighboring-ray bound avoids type distinctions along nonsupercritical paths. The two remaining disk-tangent inequalities are verified by an exact polynomial certificate over $\mathbb{Q}(\sqrt{3})$.

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1. INTRODUCTION AND PROOF OUTLINE

1.1. Notation, theorem, and canonical labeling. All distances, angles, and areas are Euclidean. A unit equilateral triangle has side length one; its open version is its interior. We write $[x, y]$ for the closed segment, $\text{conv}K$ for the convex hull, and $\text{int}K$ and ∂K for the interior and boundary. The relative interior $\text{relint}K$ is its interior within its affine span. The measure $\mathcal{H}^1$ is one-dimensional Hausdorff measure, which agrees with length on segments. We use $[t]_+ = \max\{t, 0\}$ and $\text{cross}(x, y) = x_1y_2 - x_2y_1$ for vectors in $\mathbb{R}^2$. A *trace* on a set E means its intersection with E .

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Let

$$O = (0, 0), \quad V_i = \left(\cos \frac{i\pi}{3}, \sin \frac{i\pi}{3} \right), \quad H = \text{conv}\{V_0, \dots, V_5\},$$

where indices lie in $\mathbb{Z}/6\mathbb{Z}$. Write

$$e_{i,i+1} = [V_i, V_{i+1}], \quad r_i = [O, V_i], \quad M_i = \frac{1}{2}V_i,$$

and put

$$S = \partial H \cup \bigcup_{i=0}^5 r_i.$$

We call S the *hexagon skeleton*. For $X \in \mathbb{R}^2$ and $r \geq 0$, write $\mathbb{B}(X, r) = \{Y : \|Y - X\| \leq r\}$ for the closed disk.

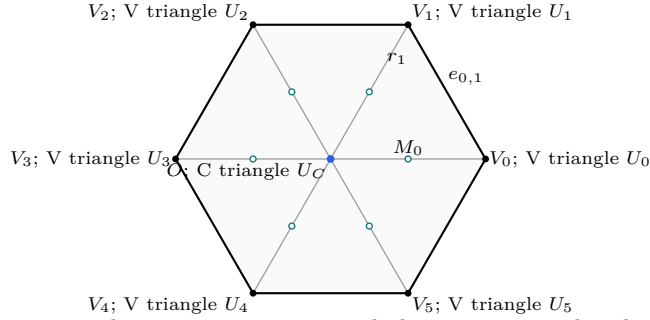


FIGURE 1. The center, vertices, radial segments, and radial midpoints of H .

For a nonempty compact set $K \subset \mathbb{R}^2$, define its *equilateral enclosure side* by

$$(1) \quad \Lambda(K) := \inf\{s > 0 : K \subset T \text{ for a closed equilateral triangle } T \text{ of side } s\}.$$

Then

$$(2) \quad \mathcal{H}^1(\partial H) = 6, \quad \mathcal{H}^1\left(\bigcup_{i=0}^5 r_i\right) = 6, \quad \mathcal{H}^1(S) = 12.$$

Theorem 1.1 (Main theorem). *The hexagon H cannot be covered by seven open equilateral triangles of side length one.*

Corollary 1.2 (Soifer's Hexagon Problem). *For every $L > 1$, the regular hexagon $H_L = LH$ cannot be covered by seven closed unit equilateral triangles.*

By lemma 2.3, a hypothetical cover has a unique labeling

$$U_C, U_0, \dots, U_5$$

such that

$$O \in U_C, \quad V_i \in U_i.$$

We call U_C the *C triangle* and U_i the *V triangle at V_i* . Set

$$T_C = \overline{U_C}, \quad T_i = \overline{U_i}.$$

A *role* means one of these labeled triangles. Statements about an arrangement of roles assume $O \in U_C$ and $V_i \in U_i$; coverage of H , S , or ∂H is an additional

hypothesis, stated when needed. A *skeleton cover* covers S , not necessarily all of H . An edge is *center-free* when it is disjoint from U_C .

1.2. **Finite classifications.** Set

$$N_E(T_C) := |\{i \in \mathbb{Z}/6\mathbb{Z} : \mathcal{H}^1(T_C \cap e_{i,i+1}) > 0\}|.$$

Proposition 2.4 gives $N_E(T_C) \in \{0, 1, 2\}$; we call these types CE0, CE1, CE2, respectively.

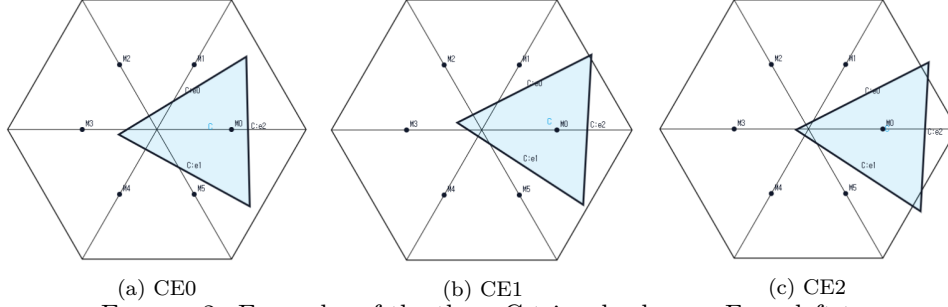


FIGURE 2. Examples of the three C triangle classes. From left to right, the C triangle has positive-length trace on zero, one, and two edges of ∂H ; point contacts do not count. The panels illustrate the classification.

For the V triangle T_i , let $o(T_i)$ be the number of its triangle vertices outside H and put

$$n(T_i) = |\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\}|,$$

with indices modulo 6. The raw pair $(3, 0)$ says that all three vertices of T_i lie outside H ; it does not say that T_i is disjoint from H , since $V_i \in \text{int}(T_i) \cap H$. Proposition 2.6 replaces every such triangle by a same-orientation translate with exactly the same trace in H , still containing V_i in its interior, and with at most two vertices outside H . We make this exact-trace normalization before defining any reach. The normalized exhaustive types used below are

$(o(T_i), n(T_i))$	established type
(1, 0)	Vd0
(2, 0)	Vd0
(1, 1)	Vd1
(1, 2)	Vd2
(2, 1)	T3-like.

The actual maximal reaches are

$$A_i := \max\{x \in [0, 1] : V_i + x(V_{i-1} - V_i) \in T_i\},$$

$$B_i := \max\{x \in [0, 1] : V_i + x(V_{i+1} - V_i) \in T_i\},$$

$$C_i := \max\{x \in [0, 1] : V_i + x(O - V_i) \in T_i\}.$$

The same numbers are the suprema of the corresponding reaches in the open triangle U_i . The V triangle T_i is *supercritical* when $A_i + B_i > 1$, and *nonsupercritical* when $A_i + B_i \leq 1$. Put

$$(3) \quad N_+ := |\{i \in \mathbb{Z}/6\mathbb{Z} : A_i + B_i > 1\}|.$$

Lowercase reaches are used only after a selection satisfying

$$0 \leq a_i \leq A_i, \quad 0 \leq b_i \leq B_i, \quad 0 \leq c_i \leq C_i.$$

They never define N_+ .

Parametrize $e_{i,i+1}$ from V_i to V_{i+1} by

$$X_i(x) := V_i + x(V_{i+1} - V_i), \quad 0 \leq x \leq 1.$$

Define the *boundary gap*

$$J_i := e_{i,i+1} \setminus (U_i \cup U_{i+1}), \quad N_{\text{gap}} := |\{i : J_i \neq \emptyset\}|.$$

Lemma 2.10 gives $J_i = X_i([B_i, 1 - A_{i+1}])$ when $B_i + A_{i+1} \leq 1$, and $J_i = \emptyset$ otherwise. Equality is a singleton gap, not an overlap.

A V role has *positive adjacent-radial support*, or simply *positive support*, when $n(T_i) > 0$. Put

$$N_{\text{sp}} := |\{i : n(T_i) > 0\}|.$$

Positive-support roles are nonsupercritical, so the number of roles that are supercritical or have positive adjacent-radial support is $N_+ + N_{\text{sp}}$. The skeleton budget excludes nonzero-gap covers with $N_+ + N_{\text{sp}} \geq 3$.

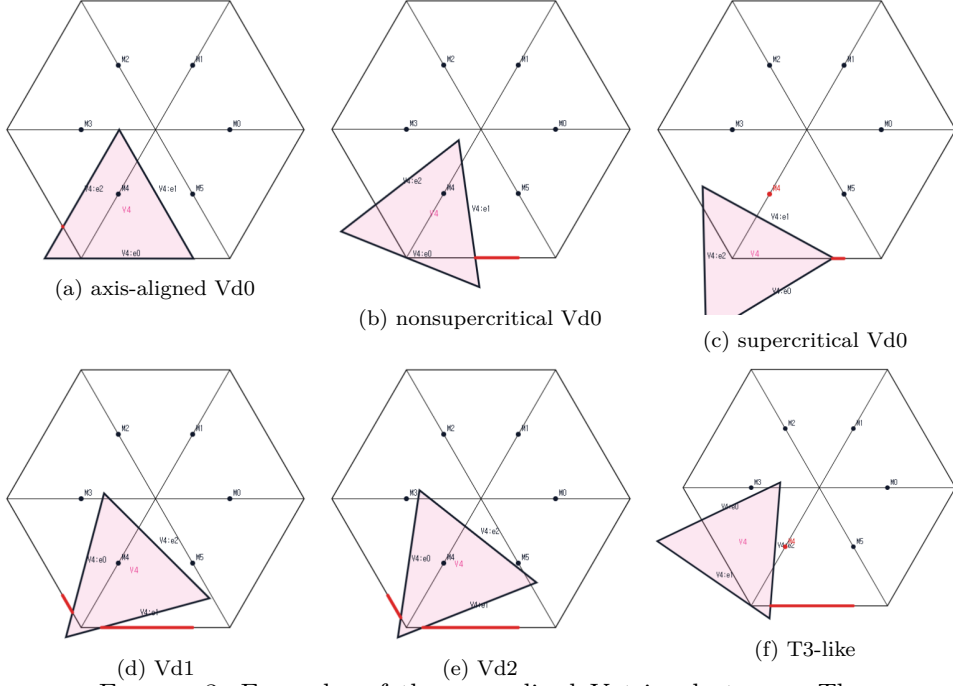


FIGURE 3. Examples of the normalized V triangle types. The top row shows an axis-aligned Vd0, a nonsupercritical Vd0 with $A_i + B_i \leq 1$, and a supercritical Vd0 with $A_i + B_i > 1$. The bottom row shows Vd1, Vd2, and T3-like. The panels illustrate the classification.

1.3. Exhaustive routing. When the C triangle has type CE1 or CE2, proposition 2.15 will show that it contains exactly one of $M_0, \dots, M_5$; this is the *C midpoint*. If it is M_k , the role at V_k is called *center-based*; a supercritical role there is *center-aligned*, and a role at another vertex is *away* or *off-center*. A V role that contains an adjacent midpoint openly is called its *supplier* or *rescuer*. A *gap-free nonsupercritical path* is a consecutive sequence of nonsupercritical V roles whose shared edges have empty gaps.

We use BC for the selected-gap six-point construction, D for the supported-midpoint four-point construction, and F for the zero-gap nine-point construction, all defined in Section 5. N0 denotes theorem 5.10, the assertion that every skeleton cover has a supercritical V role.

Zero gaps mean that the V triangles cover ∂H , and the proof splits according to N_+ . With a gap, the C triangle is CE1 or CE2; N0 and the skeleton budget leave one supercritical role and at most one positive-support role. Its position relative to the C midpoint determines the remaining obstruction.

N_{gap}	N_+	Additional condition	Closing mechanism
0	0	Arbitrary V types	Strict overlap
0	1	Arbitrary V types	Nine-point F
0	≥ 2	Arbitrary V types	Cyclic area loss
≥ 1	0	Arbitrary V types	N0, through BC
≥ 1	≥ 1	$N_+ \geq 2$ or $N_+ + N_{\text{sp}} \geq 3$	Skeleton budget
≥ 1	1	$N_{\text{sp}} \leq 1$	Midpoint supplier: BC, D, replacement, or Vd2 length

TABLE 1. Disjoint routes for the original open cover. Nonzero gaps force CE1 or CE2; both classes share the same placement reduction. Replacement preserves the skeleton and ends at N0, without preserving number of gaps.

The rows are disjoint and exhaustive: separate zero from nonzero gaps, then $N_+ = 0$, and finally the displayed high-count condition. Lemma 5.11 resolves the last row. BC uses the same selected-gap witness in CE1 and CE2, with separate candidate calculations.

1.4. Organization of the proof. Section 2 proves the structural classification, strict handoff selection, and midpoint facts. Appendix A proves the signed-center normal form used by the later arguments. Section 3 gives the complete trace-length argument. Section 4 proves the local and cyclic area-loss estimates. Section 5 gives the nonzero-gap and zero-gap enclosure obstructions. Section 6 then applies the zero-gap alternatives before the nonzero-gap midpoint-supplier reduction of Table 1 and closes the proof. Appendices A–E supply the corresponding solved optimizations; Appendix F is the exact polynomial certificate for the two mixed zero-gap intersections.

2. COMMON GEOMETRY AND STRUCTURAL REDUCTIONS

We prove the structural facts used by all three mechanisms. The coordinate calculations are in Appendix A.

2.1. Open and closed formulations.

Lemma 2.1 (Compact covering margin). *If a nonempty compact set K is covered by finitely many open unit equilateral triangles U_ν , then moving every side line inward by the same sufficiently small positive distance gives a cover by closed equilateral triangles of side less than one. In particular, $K \subset U$ implies $\Lambda(K) < 1$, with Λ defined in (1).*

Proposition 2.2 (Compactness and scaling equivalence). *The following assertions are equivalent.*

- (1) H is covered by seven open equilateral triangles of side 1.
- (2) For some $\varepsilon > 0$, H is covered by seven closed equilateral triangles of side $1 - \varepsilon$.
- (3) For some $L > 1$, $H_L := LH$ is covered by seven closed equilateral triangles of side 1.

Lemma 2.3 (Distinct triangles). *The seven triangles can be labelled so that*

$$O \in U_C, \quad V_i \in U_i \quad (0 \leq i \leq 5),$$

and these seven triangles are distinct. Consequently

$$O \in \text{int}(T_C), \quad V_i \in \text{int}(T_i).$$

2.2. C triangle and V triangle classifications. Point contacts have zero trace length, so they do not create an additional positive boundary trace in the definition of $N_E(T_C)$.

Proposition 2.4 (Exhaustive center types). *The closed C triangle T_C has exactly one of the types CE0, CE1, CE2; equivalently,*

$$N_E(T_C) \leq 2.$$

Lemma 2.5 (Local wedge). *Every original V triangle has $1 \leq o(T_i) \leq 3$. If $\mathcal{H}^1(T_i \cap r_j) > 0$ for some $j \in \{i-1, i+1\}$, at least one vertex of T_i belongs to H ; if both adjacent radial traces have positive length, at least two vertices belong to H .*

Raw roles with $(o(T_i), n(T_i)) = (3, 0)$ admit an exact-trace normalization.

Proposition 2.6 (Exact-trace normalization of raw (3,0) roles). *Let U be an original open V triangle at V_i , put $T = \overline{U}$, and suppose $(o(T), n(T)) = (3, 0)$. There is a same-orientation translate $\widehat{U}$, with closure $\widehat{T}$, such that*

$$(4) \quad V_i \in \widehat{U}, \quad (o(\widehat{T}), n(\widehat{T})) = (2, 0),$$

and

$$(5) \quad T \cap H = \widehat{T} \cap H, \quad U \cap H = \widehat{U} \cap H.$$

In particular, all three actual maximal reaches and the inequality $A_i + B_i > 1$ are unchanged.

We apply this translation simultaneously and reuse U_i, T_i for the normalized triangles.

Proposition 2.7 (Exhaustive normalized vertex types). *Every normalized V triangle has exactly one of the four types Vd0, Vd1, Vd2, T3-like in the exact dictionary of the introduction.*

The next construction is a trace majorant, not a new open covering triangle.

Proposition 2.8 (T3-like closed-trace domination). *Let T be the closure of an original T3-like V triangle at V_i . There is a same-orientation translate $\hat{T}$ such that*

$$V_i \in \text{relint}(a \text{ side of } \hat{T}), \quad T \cap H \subsetneq \hat{T} \cap H,$$

and $(o(\hat{T}), n(\hat{T})) = (2, 1)$, while

$$\text{int}(T) \cap (H \setminus \{V_i\}) \subseteq \text{int}(\hat{T}).$$

Since V_i becomes a boundary point, $\hat{T}$ is only a closed-trace majorant.

2.3. Initial reaches and boundary gaps.

Lemma 2.9 (Initial reach segments). *On the three segments from V_i toward V_{i-1}, V_{i+1}, O , the closed parameter sets are respectively*

$$[0, A_i], \quad [0, B_i], \quad [0, C_i],$$

and the open parameter sets are

$$[0, A_i), \quad [0, B_i), \quad [0, C_i).$$

Thus the closed maxima equal the corresponding open-triangle suprema.

Lemma 2.10 (Exhaustiveness of the gap split). *One has*

$$J_i = \begin{cases} X_i([B_i, 1 - A_{i+1}]), & B_i + A_{i+1} \leq 1, \\ \emptyset, & B_i + A_{i+1} > 1. \end{cases}$$

Every nonempty gap, including a singleton, is missed by all open V triangles. Under a hypothetical cover it lies in U_C and creates a positive-length C triangle trace. Consequently a closed C triangle T_C of type CE0, CE1, CE2 permits respectively at most 0, 1, 2 gaps, and

$$N_{\text{gap}} = 0 \iff U_0, \dots, U_5 \text{ cover } \partial H.$$

Lemma 2.11 (T3-like V triangles are nonsupercritical). *Every original T3-like V triangle satisfies*

$$(6) \quad A_i + B_i \leq 1.$$

Moreover $M_i \notin T_i$. In particular, a T3-like role is not counted by N_+ and cannot supply its own midpoint.

2.4. Strict handoffs and midpoint forcing.

Proposition 2.12 (Strict boundary handoffs). *Assume that $U_0, \dots, U_5$ cover ∂H . There are choices*

$$\xi_i \in (1 - A_{i+1}, B_i), \quad a_i := 1 - \xi_{i-1}, \quad b_i := \xi_i,$$

such that

$$0 < a_i < A_i, \quad 0 < b_i < B_i, \quad a_i^2 + a_i b_i + b_i^2 < 1.$$

The selected anchors lie in U_i . If exactly one actual V triangle is supercritical, every strict selection has that same unique selected supercritical index. If at least

two actual V triangles are supercritical, some simultaneous strict selection retains at least two.

Lemma 2.13 (Boundary deficit and overlap identities). *For the actual reaches put $s_i = 1 - A_i - B_i$ and $\omega_i = B_i + A_{i+1} - 1$. Then*

$$A_{i+1} - A_i = s_i + \omega_i, \quad B_i - B_{i+1} = s_{i+1} + \omega_i.$$

On a nonsupercritical gap-free path the A_i increase and the B_i decrease strictly. Weak handoffs give the weak version. Around the full cycle, $\sum_i \omega_i = -\sum_i s_i$.

Lemma 2.14 (Self-midpoint obstruction). *Let u, v be unit vectors meeting at 120° , and let $a, b \in [0, 1]$. If a closed unit equilateral triangle contains*

$$0, \quad au, \quad bv, \quad \frac{u+v}{2},$$

then $a + b \leq 1$. Consequently

$$M_i \in T_i \implies A_i + B_i \leq 1.$$

If the boundary points and midpoint are contained with positive margin, the conclusion is strict.

Proposition 2.15 (CE1/CE2 exactly-one-midpoint property). *If $O \in \text{int}(T_C)$ and T_C is CE1 or CE2, then*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1.$$

This midpoint belongs to U_C . Both possible positive boundary traces are on the two hexagon edges incident to the same vertex.

The interior hypothesis is essential: $\text{conv}\{O, V_0, V_1\}$ has a boundary edge overlap and contains both M_0 and M_1 , but O is not interior to it.

3. TRACE-LENGTH MECHANISM

For a closed triangle T and a rectifiable target E , define

$$L_E(T) := \mathcal{H}^1(T \cap E).$$

Lemma 3.1 (Connected open-cover budget). *Let X be compact and connected, with a finite Borel measure μ positive on every nonempty relatively open subset. Let $U_1, \dots, U_m$ be open sets covering X , put $T_j = \overline{U_j}$, and assume at least two of the relative pieces $U_j \cap X$ are nonempty. Then*

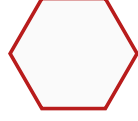
$$\mu(X) < \sum_j \mu(U_j \cap X) \leq \sum_j \mu(T_j \cap X).$$

Consequently closed-piece upper bounds whose sum is at most $\mu(X)$ already contradict the open cover.

For $X = \partial H$ or S , use length; for $X = H$, use area. Each measure has the stated full-support property. The six V roles meet each target in positive measure; the C role does so on H and S , but need not meet ∂H . In particular,

$$(7) \quad \mathcal{H}^1(E) < L_E(T_C) + \sum_{i=0}^5 L_E(T_i), \quad E \in \{\partial H, S\},$$

whenever these targets are covered. Empty center boundary traces are simply omitted from the collection. The local geometric caps remain separate from this common final budget argument.



∂H : length 6



S : length 12

the full-skeleton estimate is used only under its stated branch hypotheses

FIGURE 4. The perimeter and full-skeleton targets. The skeleton is used only under the hypotheses of the corresponding trace budget.

3.1. Skeleton bounds.

Lemma 3.2 (Center skeleton cap). *If $N_E(T_C) \in \{1, 2\}$ and*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1,$$

then

$$L_S(T_C) < \frac{3}{2}.$$

Lemma 3.3 (Positive-support skeleton cap). *Let T_i be the closure of an original open V triangle. If*

$$\exists j \in \{i-1, i+1\} \quad \mathcal{H}^1(T_i \cap r_j) > 0,$$

then

$$L_S(T_i) < \frac{3}{2}.$$

Lemma 3.4 (No-support skeleton cap). *If*

$$A_i + B_i \leq 1,$$

$$\mathcal{H}^1(T_i \cap r_{i-1}) = 0, \quad \mathcal{H}^1(T_i \cap r_{i+1}) = 0,$$

then

$$L_S(T_i) \leq 2.$$

Lemma 3.5 (Supercritical skeleton cap). *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Lemma 3.6 (Positive-support rescuer). *Assume that the open roles cover S , that $N_E(T_C) \in \{1, 2\}$, and, after symmetry,*

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

If $N_+ \geq 2$, then for a supercritical index $s \neq 0$ there is $j \in \{s-1, s+1\}$ such that

$$M_s \in U_j, \quad \mathcal{H}^1(T_j \cap r_s) > 0.$$

Theorem 3.7 (Direct CE1/CE2 skeleton count). *If*

$$\begin{aligned} N_E(T_C) &\in \{1, 2\}, \\ |T_C \cap \{M_0, \dots, M_5\}| &= 1, \end{aligned}$$

and

$$(8) \quad N_+ + N_{\text{sp}} \geq 3,$$

then the seven triangles do not cover S .

3.2. Boundary trace bounds. The following bounds suffice for the perimeter exit.

Theorem 3.8 (Boundary trace bounds). *For closures of the original open roles,*

$$L_{\partial H}(T_C) = 0 \quad (\text{CE0}), \quad L_{\partial H}(T_C) < \frac{1}{2} \quad (\text{CE1 or CE2}),$$

and

$$(9) \quad L_{\partial H}(T_i) = A_i + B_i \leq \frac{2}{\sqrt{3}}.$$

A nonsupercritical role contributes at most 1; a Vd1 or Vd2 role contributes less than 1/2.

Proposition 3.9 (Vd2 adjacent-midpoint branch). *Suppose the C triangle is CE1 or CE2 and $N_+ = 1$. If a Vd2 role contains an adjacent midpoint, the perimeter is not covered.*

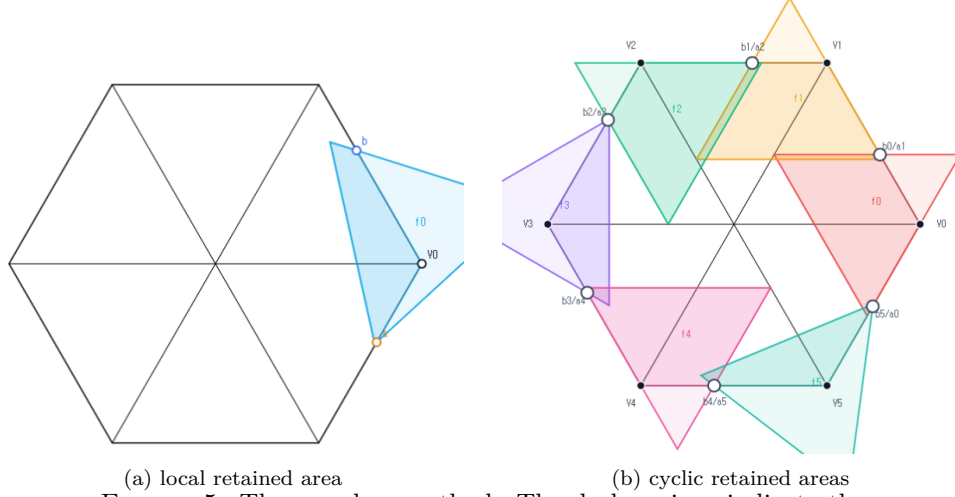
4. AREA-LOSS ESTIMATE

The area-loss method records how much of each V triangle must lie outside H . The local estimate applies to every V triangle. A final cyclic argument adds these losses without imposing a center type.

Normalize area by $A_\Delta = \sqrt{3}/4$ and write

$$\mathcal{L}(T) := \frac{\text{area}(T \setminus H)}{A_\Delta}.$$

The hexagon has normalized area six, and a unit equilateral triangle has normalized area one.



(a) local retained area
 (b) cyclic retained areas
 FIGURE 5. The area-loss method. The dark regions indicate the portions retained in H ; the loss $\mathcal{L}(T)$ is the area outside H .

4.1. Local area loss.

Theorem 4.1 (Local square loss). *Let T be a closed unit equilateral triangle containing V_i , $V_i + a(V_{i-1} - V_i)$, and $V_i + b(V_{i+1} - V_i)$, where $a, b \in [0, 1]$. Then*

$$(10) \quad \mathcal{L}(T) \geq \min\{a, b\}^2,$$

and, if $a + b > 1$,

$$(11) \quad \mathcal{L}(T) \geq \max\{a, b\}^2.$$

4.2. Cyclic accumulation. Assume here that the six V triangles cover ∂H , and choose strict handoffs ξ_i, a_i, b_i as in proposition 2.12. An *ascent* at index i means $\xi_{i-1} < \xi_i$, equivalently $a_i + b_i > 1$.

Lemma 4.2 (Cyclic area-loss certificate). *The six actual V triangles satisfy*

$$|\{i \in \mathbb{Z}/6\mathbb{Z} : a_i + b_i > 1\}| \geq 2 \implies \sum_{i=0}^5 \mathcal{L}(T_i) > 1.$$

Proposition 4.3 (Branch closed by area). *The branch $N_{\text{gap}} = 0$, $N_+ \geq 2$ is impossible.*

5. FINITE-ENCLOSURE OBSTRUCTIONS

The finite-enclosure method finds compact sets that the V triangles miss. Under a hypothetical cover those sets lie in the open C triangle, while their least equilateral enclosure has side at least one.

Put

$$h := \frac{\sqrt{3}}{2}.$$

The closed-disk notation and Λ are defined in Section 1. By lemma 2.1, a compact set contained in an open unit triangle has enclosure side below one. Conversely, $\Lambda(K) < 1$ gives such an open unit enclosure by slightly enlarging a smaller closed

enclosure. Thus it is enough to exclude every open unit candidate containing the same fixed set K .

5.1. Local capacities and radial witnesses. Normalize at V_0 and define the four-anchor hull

$$K(a, b, c) := \text{conv}\{V_0, X_5(1-a), X_0(b), (1-c)V_0\},$$

$$\mathcal{A} := \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \leq 1\}.$$

For

$$0 \leq a, b \leq 1, \quad a^2 + ab + b^2 \leq 1,$$

put

$$c_{\max}(a, b) := \max\{c \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

For $0 \leq a, c \leq 1$, also put

$$M_c(a) := \max\{b \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

In particular, the *diameter envelope* is

$$(12) \quad M_0(x) = \frac{-x + \sqrt{4 - 3x^2}}{2} \quad (0 \leq x \leq 1).$$

Here M_0 without an argument denotes the radial midpoint, whereas $M_0(x)$ denotes this scalar function. For later rescuer estimates define

$$(13) \quad M_c^{\sup} := \frac{c + \sqrt{c^2 - 8c + 4}}{2} \quad (0 \leq c \leq 1/2).$$

The value at $c = 1/2$ is a continuous endpoint convention; no supercritical triangle has that radial demand. The bound associated with this envelope is proved in lemma A.13. The exact admissible set and radial envelope are proved in proposition A.11.

The neighboring-arm capacities are the variational quantities

$$C_+(a, b) := \max \left\{ c \in [0, 1] : \begin{array}{l} \exists T \text{ [} T \text{ is a closed unit equilateral triangle,} \\ \{V_0, X_5(1-a), X_0(b), (1-c)V_1\} \subseteq T \end{array} \right\},$$

$$C_-(a, b) := C_+(b, a), \quad 0 \leq a, b \leq 1, \quad a + b \leq 1.$$

Only a bound, not the full piecewise neighboring capacity, is needed below. The direct four-caliper proof is lemma D.1.

Lemma 5.1 (Common-pair domination). *If*

$$p, q \geq 0, \quad p + q \leq 1,$$

then

$$C_+(p, q), C_-(p, q) \leq 1 - \min\{p, q\} \leq c_{\max}(p, q).$$

Definition 5.2 (Fixed total radial endpoint). Parametrize r_i by $Z_i(c) = (1-c)V_i$, measured from V_i toward O . Let $u_{j \rightarrow i}$ be the largest parameter in the actual positive trace of T_j on r_i for $j = i-1, i+1$, with value zero when no such positive trace exists. Put

$$\gamma_i := \max\{C_i, u_{i-1 \rightarrow i}, u_{i+1 \rightarrow i}\}, \quad d_i := 1 - \gamma_i, \quad \widehat{P}_i := d_i V_i.$$

Thus $\widehat{P}_i$ is the innermost endpoint reached by any local V triangle on r_i , rather than just the endpoint of the own V triangle. This distinction is essential when a nonsupercritical V triangle supports an adjacent radial segment.

Lemma 5.3 (Fixed radial forcing and demand recovery). *One has $0 < d_i < 1$ and $\widehat{P}_i \notin \bigcup_j U_j$. Under skeleton coverage, $\widehat{P}_i \in U_C$. More generally, for every $\gamma_i \leq \Gamma \leq 1$, the point $(1 - \Gamma)V_i$ is missed by every open V triangle and is center-forced under skeleton coverage. If an arbitrary open unit candidate U' contains O and $\widehat{P}_i$, and its exit on r_i is d'_i measured from O , then $\gamma_i > 1 - d'_i$. If both neighboring endpoints are at most $1 - d'_i$, it follows that*

$$C_i > 1 - d'_i.$$

It is sufficient that each neighboring V triangle be nonsupercritical with both actual boundary reaches greater than d'_i .

Let selected lower bounds satisfy

$$\begin{aligned} 0 \leq a_j \leq A_j, \quad 0 \leq b_j \leq B_j \quad (0 \leq j \leq 5), \\ a_j^2 + a_j b_j + b_j^2 \leq 1 \quad (0 \leq j \leq 5). \end{aligned}$$

A neighboring term is used only when the actual V triangle has the corresponding positive radial trace:

$$\begin{aligned} \mathcal{H}^1(T_{i-1} \cap r_i) > 0 &\implies a_{i-1} + b_{i-1} \leq 1, \\ \mathcal{H}^1(T_{i+1} \cap r_i) > 0 &\implies a_{i+1} + b_{i+1} \leq 1. \end{aligned}$$

Define

$$\begin{aligned} \Gamma_i := \max \left(\{c_{\max}(a_i, b_i)\} \cup \{C_+(a_{i-1}, b_{i-1}) : \mathcal{H}^1(T_{i-1} \cap r_i) > 0\} \right. \\ \left. \cup \{C_-(a_{i+1}, b_{i+1}) : \mathcal{H}^1(T_{i+1} \cap r_i) > 0\} \right), \quad D_i := (1 - \Gamma_i)V_i. \end{aligned}$$

The points D_i are the radial witnesses used below.

Lemma 5.4 (Type-aware radial forcing). *For every i ,*

$$D_i \in r_i \subseteq S, \quad D_i \notin \bigcup_{j=0}^5 U_j.$$

Consequently,

$$S \subseteq U_C \cup \bigcup_{j=0}^5 U_j \implies D_i \in U_C.$$

Corollary 5.5 (Uniform common-pair radial forcing). *Suppose $p, q \geq 0$, $p + q \leq 1$, and*

$$p \leq a_i \leq A_i, \quad q \leq b_i \leq B_i \quad (0 \leq i \leq 5), \quad 0 < c_* := c_{\max}(p, q) < 1.$$

Then $(1 - c_)V_i \notin \bigcup_{j=0}^5 U_j$ for every i , without a restriction on the normalized V types. Under a hypothetical cover of H ,*

$$(1 - c_*)V_i \in U_C \quad (0 \leq i \leq 5), \quad \mathbb{B}(O, h(1 - c_*)) \subset U_C.$$

5.2. Fixed witnesses and candidate enclosures. Each witness set is fixed from the original V triangles before an arbitrary enclosing candidate is chosen. The candidate's center parameters are recomputed, but its witness points are not. Gap endpoints are supplied by lemma 2.10; radial witnesses use the total endpoints, including all permitted neighboring traces.

Lemma 5.6 (The origin is implicit in the path witnesses). *For $G = X_0(t)$, $0 < t < 1$, and $d_2, d_4 > 0$,*

$$G + \frac{1-t}{d_2}(d_2V_2) + \frac{1}{d_4}(d_4V_4) = 0.$$

Consequently $O \in \text{conv}\{G, d_2V_2, d_4V_4\}$.

The three constructions are summarized after they have been defined. Normalize the C midpoint to M_0 in the nonzero-gap cases, and write σ for the unique supercritical index when $N_+ = 1$. Counts are upper bounds, allowing coincident gap endpoints.

Replacement is a separate skeleton-preserving operation, not a fourth witness construction. BC handles the center-aligned case. For an off-center supercritical role, its midpoint supplier leads to D, replacement, or the Vd2 perimeter obstruction.

5.3. BC: one selected gap and one six-point construction. Normalize the center midpoint to M_0 and choose any actual gap, reflected to J_0 . The other incident edge $e_{5,0}$ may or may not contain a gap. Only the four middle edges $e_{1,2}, \dots, e_{4,5}$ must be gap-free. The following transfer supplies the tail bound needed by the path argument.

Lemma 5.7 (Shared-gap-anchor boundary transfer). *If the original open roles cover ∂H and J_0 is an actual gap, then, independently of the gap status of $e_{5,0}$,*

$$B_5 > 1 - M_0(B_0) > B_0/2.$$

Here $M_0(z)$ is the scalar diameter envelope.

In the pure enclosure statement below the tail bound $B_5 \geq B_0/2$ is explicit. In the covering application, the preceding lemma establishes it for the original roles before an arbitrary enclosing candidate is chosen. We do not assume that the candidate also covers the discarded gap.

Define

$$K_{BC} := \{M_0, X_0(B_0), X_0(1 - A_1), \widehat{P}_2, \widehat{P}_3, \widehat{P}_4\}.$$

The midpoint is a structural anchor belonging to the original open C triangle. The other points are missed by the V triangles. The origin is implicit by lemma 5.6; the construction uses at most six points even though a candidate must also contain O .

Theorem 5.8 (Unified six-point selected-gap enclosure). *Suppose J_0 is an actual gap, $A_i + B_i \leq 1$ for $1 \leq i \leq 5$, and*

$$B_i + A_{i+1} > 1 \quad (1 \leq i \leq 4), \quad B_5 \geq B_0/2.$$

Then $\Lambda(K_{BC}) \geq 1$. Under skeleton coverage with distinguished C midpoint M_0 , the fixed set K_{BC} lies in U_C . There is no condition on the gap status of $e_{5,0}$ or supercriticality of T_0 , and no type restriction on the nonsupercritical path.

Theorem 5.9 (Center-aligned path obstruction). *Suppose a skeleton cover has at least one actual gap and its C triangle's unique midpoint is M_k . If every T_i with $i \neq k$ is nonsupercritical, the configuration is impossible.*

Theorem 5.10 (Every skeleton cover has a supercritical V triangle). *A skeleton cover by the seven original open roles satisfies $N_+ \geq 1$.*

Lemma 5.11 (A unique midpoint supplier). *In a nonzero-gap skeleton cover, the preceding N0 theorem and the skeleton budget leave only $N_+ = 1$, $N_{\text{sp}} \leq 1$. Write M_k for the C midpoint and σ for the supercritical index. If $\sigma \neq k$, there is a unique positive-support role T_τ , where*

$$\tau \in \{\sigma - 1, \sigma + 1\}, \quad M_\sigma \in U_\tau.$$

If T_τ is T3-like, then $\tau = k$. If $\tau \neq k$, the edge shared by T_τ, T_σ is center-free, in both CE1 and CE2.

5.4. D: a four-point supported-rescuer enclosure. Let $Y(t) = (1 - t)V_0 + tV_5 = X_5(1 - t)$. The geometric lemma below is stated independently of any V type or covering configuration.

Theorem 5.12 (Four-point rescuer geometry). *Suppose $a \geq 0$, $\varepsilon > 0$, $\beta \geq 0$, and*

$$s = a + \varepsilon \leq 1, \quad a \leq \varepsilon, \quad \beta \leq \frac{\varepsilon}{s}.$$

Then

$$\Lambda\{O, \varepsilon V_1, Y(a), Y(1 - \beta)\} \geq 1.$$

Lemma 5.13 (Scalar rescuer ratio test). *For $0 \leq x, c \leq 1/2$ and $M = (c + \sqrt{c^2 - 8c + 4})/2$,*

$$x \leq 1 - M \iff x^2 + (c - 2)x + c \geq 0.$$

For the covering application, let T_0 supply M_1 , with supported interval $[c, u]$ on r_1 , measured from V_1 toward O , and put $\varepsilon = 1 - u$ and $P_T = \varepsilon V_1$. All interval endpoints here belong to the original triangle, not to its translated majorant. The local T3-like and Vd1 calculations establish

$$A_0 + \varepsilon \leq 1, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M_c^{\text{sup}},.$$

Together with $C_1 \geq c$, the boundary path, and the forced endpoint, lemma D.11 gives the single fixed witness

$$K_D = \{O, \varepsilon V_1, Y(A_0), Y(1 - B_5)\} \subset U_C, \quad \Lambda(K_D) \geq 1.$$

The two local verifications are lemmas D.12 and D.13; the enclosure step does not distinguish one from two gaps.

Proposition 5.14 (T3-like supported-rescuer terminal). *Suppose $N_+ = 1$, there is exactly one T3-like role and no Vd1/Vd2 role, and there is at least one actual gap. Then the skeleton is not covered.*

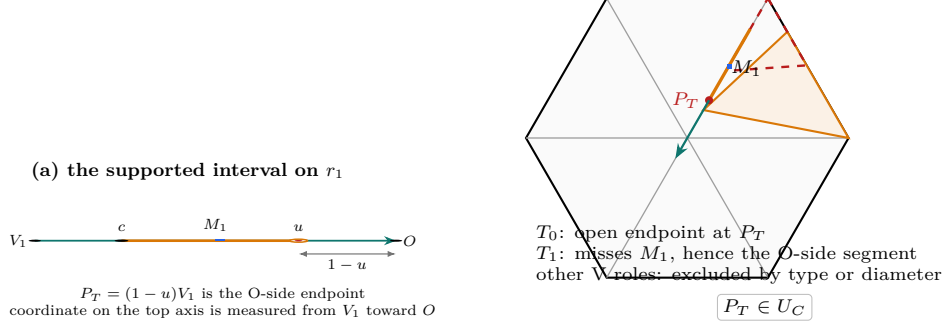
(b) why P_T is center-forced

FIGURE 6. Construction of the T3-like rescuer endpoint. The interval $[c, u]$ is parametrized from V_1 toward O , so its O-side endpoint is $P_T = (1-u)V_1$. The T3-like open role omits its endpoint, the adjacent supercritical role contains V_1 but misses M_1 and therefore misses the O-side segment by convexity, and all remaining V roles are excluded. Thus P_T is forced into the C triangle.

5.5. The remaining placement router.

Proposition 5.15 (One-Vd placement assembly). *Suppose the C triangle is CE1 or CE2, $N_+ = 1$, exactly one V triangle is Vd1 or Vd2, and there is an actual gap. The skeleton is not covered.*

5.6. The zero-gap nine-point obstruction. Assume

$$N_{\text{gap}} = 0, \quad N_+ = 1.$$

Rotate the unique supercritical role to index 4. The exact-one handoff calculation lemma E.2 gives

$$(14) \quad \xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

Define

$$(15) \quad a := 1 - \xi_3, \quad b := \xi_4, \quad p := 1 - b, \quad q := 1 - a.$$

Then

$$(16) \quad \begin{aligned} 0 < a, b < 1, \quad a &= a_4 < A_4, \quad b = b_4 < B_4, \\ a + b &> 1, \quad a^2 + ab + b^2 < 1, \\ a_i &\geq p, \quad b_i \geq q \quad (0 \leq i \leq 5). \end{aligned}$$

Define the two boundary anchors

$$P_A := V_4 + a(V_3 - V_4), \quad P_B := V_4 + b(V_5 - V_4),$$

and the feasible union

$$\mathcal{R}_{AB}(a, b) := \bigcup \left\{ T \cap H : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ V_4, P_A, P_B \in T \end{array} \right\}.$$

Put

$$c_* := c_{\max}(p, q), \quad r_* := h(1 - c_*), \quad P_i^{\text{rad}} := (1 - c_*)V_i \quad (0 \leq i \leq 5).$$

The bounds in lemma E.3 give $0 < c_* < 1$. Applying corollary 5.5 to the common pair gives

$$(17) \quad \mathbb{B}(O, r_*) \subseteq \text{conv}\{P_0^{\text{rad}}, \dots, P_5^{\text{rad}}\} \subset U_C.$$

The two circles adjacent to the supercritical frontier are

$$\mathcal{C}_- := \partial\mathbb{B}(X_2(b), 1), \quad \mathcal{C}_+ := \partial\mathbb{B}(X_5(1-a), 1).$$

Let ℓ_-, ℓ_+ be the two straight pieces of the interior frontier of $\mathcal{R}_{AB}(a, b)$, in the order of theorem E.1. Define

$$\{Q_0\} = \ell_- \cap \ell_+, \quad \{Q_-\} = \ell_- \cap \mathcal{C}_-, \quad \{Q_+\} = \ell_+ \cap \mathcal{C}_+.$$

The coordinate formulas and selected first roots are given in Appendix E, equations (58) and (62). Lemma E.4 proves that these intersections are singletons on the indicated pieces. The nine-point set is

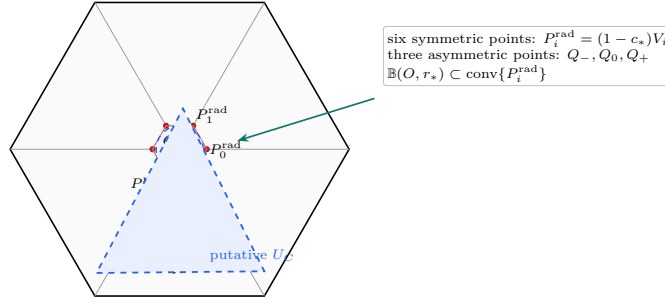
$$K_F := \{P_i^{\text{rad}} : 0 \leq i \leq 5\} \cup \{Q_-, Q_0, Q_+\}.$$

Their exact construction and exclusions in theorem E.1 and lemmas E.4 and E.5 use contained vertices and actual handoffs, not Vd0 locality, and give

$$(18) \quad Q_-, Q_0, Q_+ \in U_C.$$

For the enclosure calculation use the compact comparison set

$$(19) \quad K_{\text{wit}}(a, b) := \mathbb{B}(O, r_*) \cup \{Q_-, Q_0, Q_+\} \subset U_C.$$



$K_{\text{wit}} = \mathbb{B}(O, r_*) \cup \{Q_-, Q_0, Q_+\} \subset U_C$ under a hypothetical cover

FIGURE 7. The type-independent zero-gap nine-point witness. The six symmetric radial points force a centered disk into the convex C triangle, while the three asymmetric frontier points supply the missing directional support. Marker shapes separate the two constructions: circles are radial witnesses and triangles are strict-supercritical AB -frontier witnesses. The displayed placement is illustrative; the point definitions are the exact ones in the preceding lemmas.

Theorem 5.16 (Witness enclosure). *For $0 < a, b < 1$, $a+b > 1$, and $a^2+ab+b^2 < 1$, the comparison set $K_{\text{wit}}(a, b)$ satisfies*

$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

Theorem 5.17 (Zero-gap obstruction). *There is no hypothetical cover with*

$$N_{\text{gap}} = 0, \quad N_+ = 1.$$

5.7. Summary of the fixed witnesses.

Table 2: Fixed finite-enclosure recipes. Ordinary nonsupercritical V triangles are not subdivided by type.

Conditions	Fixed witness set	Disk and count
BC: at least one incident gap; $T_1, \dots, T_5$ nonsupercritical	$\{M_0, X_0(B_0), X_0(1 - A_1), \widehat{P}_2, \widehat{P}_3, \widehat{P}_4\}$	No disk; at most 6
D: center-based T3-like or Vd1 rescuer; one or two gaps	$\{O, \varepsilon V_1, Y(A_0), Y(1 - B_5)\}$	No disk; at most 4
F: zero gaps, $N_+ = 1$	The nine-point set K_F	Disk; at most 9

6. COMPLETION OF THE PROOF

APPENDIX A. STRUCTURAL, SHARED-LOCAL, AND SIGNED-CENTER OPTIMIZATION

This appendix solves the coordinate problems used by the geometric interfaces in Section 2. It also proves the shared equilateral-enclosure calculus used by the later optimization appendices. Each calculation is attached to the body statement it supplies.

A.1. Center and V triangle incidence feasibility.

Lemma A.1 (Separation of nonadjacent boundary edges). *Relative-interior points on two nonadjacent edges of H are more than unit distance apart.*

Lemma A.2 (Shared corner chart and wedge reduction). *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \quad W = \{(x, y) : x \geq 0, y \geq 0\}.$$

If a closed unit equilateral triangle T contains V_0 , then

$$T \cap H = T \cap W.$$

Normalized Euclidean area is twice the corresponding (x, y) -coordinate area.

Lemma A.3 (Local wedge calculation). *The incidence conclusions of Lemma 2.5 hold.*

A.2. Shared orientation charts and exact-trace normalization. Set

$$D_t := 1 - t + t^2, \quad z := \sqrt{D_t}, \quad 0 \leq t \leq 1.$$

In the shared corner chart, physical rotation through 60° is $(x, y) \mapsto (x - y, x)$. After relabelling the vertices, every orientation belongs to one of the two families

	first side vector	second side vector
I	$z^{-1}(1, t)$	$z^{-1}(1 - t, 1)$
II	$z^{-1}(t, -(1 - t))$	$z^{-1}(1, t)$

They cover successive 60° direction sectors, including their common endpoints. The following half-plane descriptions allow closed containment of the corner; strict interior containment is recorded separately. Here U, V are affine slack functions, not the open triangles U_i . In Type I write

$$(20) \quad U = \alpha + y - tx, \quad V = \beta + x - (1 - t)y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\},$$

and in Type II write

$$(21) \quad U = \alpha + (1 - t)x + ty, \quad V = \beta + tx - y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\}.$$

In both charts

$$(22) \quad V_0 \in \text{int}(T) \iff \alpha > 0, \quad \beta > 0, \quad \alpha + \beta < z.$$

The Type-II vertex in the wedge is

$$(23) \quad Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z - \alpha) + (1 - t)\beta}{D_t} \right).$$

Lemma A.4 (Raw $(3, 0)$ trace-normalization calculation). *The translation asserted in Proposition 2.6 exists and preserves both the open and closed traces in H .*

A.3. T3-like trace-majorant feasibility. The calculation uses the local wedge and the two orientation charts just introduced. Its output is a closed majorant, not a replacement open triangle.

Lemma A.5 (T3-like translation calculation). *The closed-trace majorant in Proposition 2.8 exists. In the calculation chart, every original T3-like triangle also satisfies $A_i + B_i \leq 1$ and $M_i \notin T_i$.*

A.4. Strict-handoff feasibility.

Lemma A.6 (Cyclic strict-handoff selection). *Under the hypotheses of Proposition 2.12, all six overlap intervals are nonempty, and the selectors can preserve the stated one- or two-supercritical-index alternatives.*

A.5. Shared equilateral-enclosure optimization. Let R denote counterclockwise rotation through $2\pi/3$. For a nonempty compact $K \subset \mathbb{R}^2$, put

$$h_K(n) := \max_{x \in K} \langle x, n \rangle.$$

Proposition A.7 (Equilateral enclosure gauge). *The least side length $\Lambda(K)$ of a closed equilateral triangle containing K is*

$$(24) \quad \Lambda(K) = \frac{2}{\sqrt{3}} \min_{\|n\|=1} \sum_{j=0}^2 h_K(R^j n).$$

It is monotone under inclusion, translation invariant, and positively homogeneous.

Lemma A.8 (Support-cell rotation). *A least equilateral enclosure of a nondegenerate polygon can be chosen with one side containing a polygon edge. For a centered disk together with finitely many points, it suffices to test point–point and point–disk ties, or a disk-only orientation. Here a support cell is an open angular interval on which the maximizing point or disk is fixed in each of the three support directions; a tie is an endpoint where a maximizer changes.*

A.5.1. Finite caliper criterion. Let J denote counterclockwise rotation through $\pi/2$. For a nonempty finite set F define

$$(25) \quad W_F(N) := \sum_{k=0}^2 \max_{z \in F} \langle z, R^k N \rangle.$$

Theorem A.9 (Finite caliper certificate). *If F has at least two distinct points, then F is contained in a closed unit equilateral triangle if and only if there are distinct $p, q \in F$ and $\varepsilon \in \{-1, 1\}$ such that*

$$(26) \quad W_F(\varepsilon J(q - p)) \leq \frac{\sqrt{3}}{2} \|p - q\|.$$

A singleton is contained in equilateral triangles of arbitrarily small positive side length.

Proposition A.10 (Disk–finite-set calipers). *Let $\rho > 0$, $P = \{p_1, \dots, p_m\} \subset \mathbb{R}^2$, and $K = \text{conv}(\mathbb{B}(O, \rho) \cup P)$. Use the rotation J defined above. Unless the disk is the active support in all three directions at a minimizing orientation, a minimizing normal belongs to the finite list*

$$\pm \frac{J(p_k - p_i)}{\|p_k - p_i\|} \quad (p_i \neq p_k)$$

or

$$\frac{\rho p_i \pm \sqrt{\|p_i\|^2 - \rho^2} J p_i}{\|p_i\|^2} \quad (\|p_i\| \geq \rho).$$

Thus one side of a least enclosing equilateral triangle either contains two points, or is tangent to the disk and contains one point.

A.6. The local admissible set and radial envelope. Let u, v be unit vectors meeting at 120° and define

$$K(a, b, c) := \text{conv}\{0, au, bv, c(u+v)\}.$$

This is the translate of the body normalization at V_0 , with $u = V_5 - V_0$ and $v = V_1 - V_0$. The exact local admissible set is

$$(27) \quad \mathcal{A} := \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \leq 1\}.$$

This set records the three reach lower bounds that a closed unit V triangle can realize.

Proposition A.11 (Exact local admissible set). *Put*

$$s := a + b, \quad d := a^2 + ab + b^2, \quad m := \min\{a, b\}, \quad M := \max\{a, b\}, \\ q := s^4 - s^2 + ab,$$

and

$$F_L := c^4 - c^2 + mc - m^2, \\ F_T := (s^2 - 1)c^2 + Mc - M^2, \\ F_S := (m^2 - 1)c^2 + (2mM^2 + M)c + M^4 - M^2.$$

Then

$$(28) \quad \mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S,$$

where all variables lie in $[0, 1]$ and

$$\mathcal{A}_L := \{d \leq 1, s \leq 1, q \leq 0, F_L \leq 0\}, \\ \mathcal{A}_T := \{d \leq 1, s \leq 1, q \geq 0, F_T \leq 0, c \leq 2M\}, \\ \mathcal{A}_S := \{d \leq 1, s > 1, F_S \leq 0, c \leq 1/2\}.$$

The set is coordinatewise down-closed. Define its maximal radial demand by

$$c_{\max}(a, b) := \max\{c \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

Then

$$(29) \quad c_{\max}(a, b) = \begin{cases} 1, & a = b = 0, \\ C_L(m), & s \leq 1, q \leq 0, (a, b) \neq (0, 0), \\ \frac{2M}{1 + \sqrt{4s^2 - 3}}, & s \leq 1, q > 0, \\ \frac{M(2mM + 1) - M\sqrt{4d - 3}}{2(1 - m^2)}, & s > 1, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

For later variational use, the raw forward envelope retains its established notation

$$M_c(a) := \max\{b \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

No closed-form evaluation of M_c is needed in the geometric body.

A.7. Self-midpoint corollary.

Lemma A.12 (Self-midpoint admissibility corollary). *If $a + b > 1$, then*

$$\Lambda\left(\text{conv}\left\{0, au, \frac{u+v}{2}, bv\right\}\right) > 1.$$

This supplies Lemma 2.14.

Lemma A.13 (Strict-supercritical outgoing envelope). *If a V triangle is supercritical and has own-radial reach at least $c \in [0, 1/2)$, then both boundary reaches are strictly below M_c^{sup} from (13). No supercritical triangle has own-radial reach at least $1/2$.*

A.8. Signed-center optimization. The CE1 and CE2 cases share one signed cell. The parameters below are used only in the appendices; the geometric body uses the exact C triangle type and midpoint interfaces.

Proposition A.14 (Signed CE1/CE2 center normal form). *Let T_C be a closed unit equilateral triangle with $O \in \text{int}(T_C)$ and a positive-length boundary trace. Normalize such a trace to $e_{0,1}$ and use affine coordinates*

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0).$$

There are parameters

$$\begin{aligned} 0 < R < 1, \quad W &:= 1 - R, \quad E := \sqrt{1 - RW}, \\ \eta &:= 1 - E, \quad P := E(1 - E), \end{aligned}$$

and nonnegative slacks α, δ , with $k := \eta + \alpha + \delta$, such that

$$\begin{aligned} F_0 &= R + \alpha - a + Wb, \\ (30) \quad F_1 &= Rb + Wa - k, \quad T_C = \{F_0, F_1, F_2 \geq 0\}. \\ F_2 &= W + \delta - b + Ra, \end{aligned}$$

Define

$$(31) \quad \Delta_R := P - \alpha - W\delta, \quad \Delta_L := P - R\alpha - \delta.$$

Then

$$(32) \quad T_C \cap e_{0,1} = \left[\frac{k}{R}, W + \delta\right], \quad \mathcal{H}^1(T_C \cap e_{0,1}) = \frac{\Delta_R}{R},$$

and the possible companion trace is

$$(33) \quad T_C \cap e_{5,0} = \left[\frac{k}{W}, R + \alpha\right] \quad \text{when } \Delta_L > 0, \quad \mathcal{H}^1(T_C \cap e_{5,0}) = \frac{[\Delta_L]_+}{W}.$$

The normalized trace requires $\Delta_R > 0$, and

$$(34) \quad T_C \text{ is CE1} \iff \Delta_L \leq 0, \quad T_C \text{ is CE2} \iff \Delta_L > 0.$$

The six radial exits from O are

$$(35) \quad \begin{aligned} d_0^C &= E - \alpha - \delta, & d_1^C &= \frac{\delta}{R}, & d_2^C &= \delta, \\ d_3^C &= \min \left\{ \frac{\alpha}{R}, \frac{\delta}{W} \right\}, & d_4^C &= \alpha, & d_5^C &= \frac{\alpha}{W}. \end{aligned}$$

Moreover,

$$(36) \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

For the closure of an original open C triangle, $\alpha, \delta > 0$.

Lemma A.15 (CE2 total slack). *In the CE2 sign range, with $T := \alpha + \delta$,*

$$(37) \quad T < \frac{\eta}{E}.$$

Corollary A.16 (Common center-boundary formula). *The full center-boundary contribution is*

$$(38) \quad L_{\partial H}(T_C) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}.$$

In CE1,

$$L_{\partial H}(T_C) \leq P \leq \frac{\sqrt{3}}{2} - \frac{3}{4},$$

and in CE2,

$$L_{\partial H}(T_C) = \frac{E}{1+E} - \frac{E^2(\alpha + \delta)}{RW} < \frac{1}{2}.$$

Lemma A.17 (Adjacent-edge diameter transfer). *For $0 \leq q \leq 1$, the zero-radial forward envelope is*

$$M_0(q) = \frac{-q + \sqrt{4 - 3q^2}}{2}.$$

Here M_0 without an argument remains the radial midpoint. If a unit equilateral triangle reaches parameters q, b on adjacent boundary edges, then $b \leq M_0(q)$. The function is strictly decreasing and, for $q > 0$,

$$M_0(q) < 1 - \frac{q}{2}, \quad 1 - M_0(q) > \frac{q}{2}.$$

APPENDIX B. TRACE-LENGTH OPTIMIZATION

This appendix evaluates the trace quantities whose exact interfaces appear in Section 3. The center and V triangle coordinates are used only here; the body retains the geometric hypotheses and final bounds.

B.1. Vd1 and Vd2 corner optimization. Use temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0),$$

so $V_0 = (0, 0)$, $V_5 = (1, 0)$, $V_1 = (0, 1)$, $O = (1, 1)$, and $\|(x, y)\|^2 = x^2 + y^2 - xy$.

Proposition B.1 (Vd1/Vd2 corner normal form). *Let T be the closure of an original Vd1 or Vd2 triangle at V_0 , and let A_0, B_0 be its exact reaches toward V_5, V_1 . There is a unique $t > 0$ such that, with $d = \sqrt{t^2 + t + 1}$,*

$$(39) \quad T = \left\{ (x, y) : \begin{aligned} x - (t+1)y &\leq A_0, \\ ty - (t+1)x &\leq tB_0, \\ tx + y &\leq d - A_0 - tB_0 \end{aligned} \right\}.$$

The raw traces, parametrized from the indicated outer vertex, are

$$(40) \quad \begin{array}{l} r_0 \\ r_1 \\ r_5 \end{array} \left| \begin{array}{l} 0 \leq s \leq \frac{d - A_0 - tB_0}{t+1} \\ \frac{t(1-B_0)}{t+1} \leq s \leq \frac{d - A_0 - tB_0 - 1}{t} \\ \frac{1-A_0}{t+1} \leq s \leq d - A_0 - tB_0 - t. \end{array} \right.$$

Intersecting with $[0, 1]$ gives the actual traces, and

$$(41) \quad A_0 + B_0 < \frac{1}{2}.$$

Lemma B.2 (Orientation of a forward Vd1 supplier). *In proposition B.1, if a Vd1 role contains M_1 openly, then $t \geq 1$.*

Lemma B.3 (Vd2 adjacent-midpoint cap). *If an original open V triangle has Vd2 closure T_i and*

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \emptyset,$$

then its exact boundary reaches satisfy

$$A_i + B_i < \frac{1}{3}.$$

B.2. Conditional skeleton caps.

Lemma B.4 (Center-skeleton calculation). *Under the hypotheses of Lemma 3.2, one has $L_S(T_C) < 3/2$.*

For the remaining cap, reflect the local directions if necessary so that $0 \leq A_i \leq B_i \leq 1$. The actual reach triple of a supercritical V triangle lies in the selected cell

$$(42) \quad \begin{aligned} A_i^2 + A_i B_i + B_i^2 &\leq 1, \\ C_i &\leq \frac{1}{2}, \\ (A_i^2 - 1)C_i^2 + (2A_i B_i^2 + B_i)C_i + (B_i^4 - B_i^2) &\leq 0, \end{aligned}$$

where C_i belongs to the connected component containing 0 in the corresponding one-dimensional fiber. This is Cell S of Proposition A.11. The selector is essential: the quadratic is negative at $C_i = 0$, positive at $C_i = 1/2$, and opens downward.

Lemma B.5 (Supercritical skeleton calculation). *Every V triangle with $A_i + B_i > 1$ has $L_S(T_i) < 3/2$.*

APPENDIX C. AREA-LOSS OPTIMIZATION

This appendix proves the pointwise local loss bound of theorem 4.1. The cyclic accumulation is proved in the body.

C.1. Local square-loss optimization.

Lemma C.1 (Solved local square-loss optimization). *Every triangle satisfying the hypotheses of theorem 4.1 obeys equations (10) and (11).*

APPENDIX D. NONZERO-GAP FINITE-ENCLOSURE OPTIMIZATION

This appendix proves the selected-gap and supported-rescuer enclosure statements and the two-vertex replacement. The witnesses are fixed in Section 5; candidate parameters vary only in their enclosure proofs.

D.1. Direct neighboring-ray domination. The capacities keep their variational definitions. The following proof establishes the bound needed by all active witnesses without evaluating the cubic plateau and radical branches of the full function.

Lemma D.1 (Sharp diagonal neighboring capacity). *For $0 \leq m \leq 1/2$, one has $C_+(m, m) = C_-(m, m) = 1 - m$.*

Lemma D.2 (Solved common-pair comparison). *The inequalities in lemma 5.1 hold.*

D.2. Exact direct output formulas. The following outgoing capacities follow from Proposition A.11 and supply the CE1 return.

For a nonsupercritical triangle with incoming reach at least a and radial reach at least c , let

$$\mathcal{B}_c(a) = \max\{b : (a, b, c) \text{ is admissible and } a + b \leq 1\}.$$

This is the outgoing capacity of a single triangle.

For $1/2 < c \leq \sqrt{3}/2$, put

$$h_c = \frac{2c}{1 + \sqrt{16c^2 - 3}}.$$

For $\sqrt{3}/2 < c < 1$, put

$$\ell(c) = \frac{c}{2} \left(1 - \sqrt{4c^2 - 3}\right), \quad r(c) = c - \ell(c).$$

Also set

$$\begin{aligned} \mathcal{Q}_-(a, c) &= \frac{\sqrt{a^2 - ac + c^2}}{c} - a, \\ \mathcal{Q}_+(a, c) &= \frac{2(1 - a^2)c^2}{2ac^2 + c + \sqrt{(2ac^2 + c)^2 - 4(1 - c^2)(1 - a^2)c^2}}. \end{aligned}$$

Proposition D.3 (Four direct high-radial outputs). *For $1/2 < c \leq \sqrt{3}/2$,*

$$\mathcal{B}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ \mathcal{Q}_+(a, c), & 1 - c < a < h_c, \\ h_c, & a = h_c, \\ \mathcal{Q}_-(a, c), & h_c < a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

For $\sqrt{3}/2 < c < 1$,

$$\mathcal{B}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ \mathcal{Q}_+(a, c), & 1 - c < a \leq \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ \mathcal{Q}_-(a, c), & r(c) \leq a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

At $a = \ell(c)$, the selected value is $r(c)$; immediately to the right it is $\ell(c)$.

D.3. High-radial and selected-chord calculations. For

$$0 < e < 1 - \frac{\sqrt{3}}{2},$$

define the selected low root

$$(43) \quad \epsilon(e) = \frac{1-e}{2} \left(1 - \sqrt{4(1-e)^2 - 3} \right).$$

It satisfies

$$(44) \quad e < \epsilon(e), \quad \epsilon(e) < \frac{2e}{1-2e},$$

and, for $e \leq 1/8$,

$$(45) \quad \epsilon(e) \leq 2e + 5e^2.$$

These follow by substituting the proposed upper bounds in the selected quadratic and checking its sign.

Lemma D.4 (Direct high-radial threshold). *Let $0 < e < 1 - \sqrt{3}/2$. A nonsupercritical V triangle has incoming boundary reach A and outgoing boundary reach B . If*

$$C \geq 1 - e, \quad A > \epsilon(e),$$

then

$$\boxed{B \leq \epsilon(e)}.$$

Lemma D.5 (Selected-arc chord bounds). *Put $c = 1 - e$, with $0 < e < 1/4$, and define $q_*(p) = 1 - \mathcal{Q}_+(p, c)$ on the selected branch. Its right endpoint is*

$$(p_e, q_e) = \begin{cases} (\epsilon(e), e + \epsilon(e)), & c > h, \\ (h_c, 1 - h_c), & c \leq h, \end{cases} \quad h = \sqrt{3}/2.$$

The function q_* is increasing and concave on $[e, p_e]$, with $q_*(e) = e$. Thus, for $e < p \leq p_e$,

$$q_*(p) \geq p + \lambda_e(p - e), \quad \lambda_e = \frac{q_e - p_e}{p_e - e}.$$

One has $\lambda_e > 1 - 4e$ when $0 < e < 1 - h$, and $\lambda_e > 1 - 5e$ throughout $0 < e < 1/4$. A following actual incoming reach q , forced by a weak boundary handoff, satisfies $q \geq q_*(p)$; equality is not assumed for actual reaches.

D.4. CE1 reverse-path calculation. The first lemma uses only the T_4 radial demand. The full return adds the T_3 and T_2 demands; BC establishes them before using the latter.

Use the signed center parameters of proposition A.14, with $w = 1 - R$, $A = \alpha$, and $D = \delta$. Assume

$$(46) \quad A > 0, \quad D > 0, \quad A + wD < P, \quad D + RA \geq P.$$

Put

$$(47) \quad X = R - D, \quad m = \frac{A}{R}, \quad h_0 = \frac{\eta + A + D}{2R}.$$

Lemma D.6 (First CE1 return step). *Assume (46), and let $T_1, \dots, T_4$ be nonsupercritical V triangles of arbitrary types, with actual reaches satisfying*

$$(48) \quad A_1 \geq X, \quad B_i + A_{i+1} \geq 1 \ (1 \leq i \leq 3), \quad B_4 \geq h_0.$$

If $C_4 \geq 1 - A$, then the reflected T_4 state lies on the selected $\mathcal{Q}_+$ component and

$$(49) \quad B_3 > L_1, \quad L_1 = (2 - 4A)h_0 - (1 - 4A)A.$$

No own-radial demand at T_2 or T_3 is assumed.

Proposition D.7 (CE1 tail-input three-transverse obstruction). *Under the domain, path, and tail hypotheses of lemma D.6, the additional own-radial demands*

$$(50) \quad C_4 \geq 1 - A, \quad C_3 \geq 1 - m, \quad C_2 \geq 1 - D$$

are incompatible.

D.5. Fixed endpoints and arbitrary candidates.

Lemma D.8 (Total radial endpoint and maximum-to-own-demand). *The assertions of lemma 5.3 hold for arbitrary V types.*

Lemma D.9 (Fixed six-point transverse enclosure). *The set K_{BC} in theorem 5.8 has $\Lambda(K_{BC}) \geq 1$ for arbitrary V types on the nonsupercritical path.*

Lemma D.10 (Fixed four-point supported-rescuer proof). *The geometric statement of theorem 5.12 holds.*

D.6. The supported-rescuer adapter.

Lemma D.11 (Supported endpoint to four-point enclosure). *Suppose T_0 has actual boundary endpoint $a = A_0$ toward V_5 and supported interval $[c, u]$ on r_1 , measured from V_1 toward O . Put $\varepsilon = 1 - u > 0$ and $M = M_c^{\text{sup}}$. Assume*

$$\varepsilon V_1 \in U_C, \quad A_0 + \varepsilon \leq 1, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M.$$

If T_1 is uniquely supercritical with $C_1 \geq c$, and the four center-free edges from $e_{1,2}$ through $e_{4,5}$ join nonsupercritical T_2, T_3, T_4, T_5 , then the skeleton is not covered. The conclusion applies to both possible nonzero gap ranks.

D.7. T3-like supported-tail adapter.

Lemma D.12 (Solved T3-like supported-tail adapter). *Under the off-center supplier placement of proposition 5.14, the actual supported interval satisfies the hypotheses of lemma D.11.*

Lemma D.13 (Solved Vd1 supported-tail adapter). *In the center-based Vd1 supplier placement, with $M_1 \in U_0$ and T_1 uniquely supercritical, the actual corner data imply the size and ratio inequalities required by lemma D.11.*

D.8. Vd1 two-chart replacement. The replacement below uses two different vertex charts. It preserves the five skeleton pieces that can change, then invokes the all-nonsupercritical obstruction without assuming preservation of the input gap rank.

Lemma D.14 (Half-square admissibility). *If $(x, y, z) \in \mathcal{A}$ satisfies*

$$x \geq \frac{1}{2}, \quad 0 \leq z \leq \frac{1}{2},$$

then

$$y \leq 1 - z.$$

Lemma D.15 (Two-vertex replacement from scalar margins). *Consider two adjacent V roles. Only the five pieces*

$$e_{5,0}, \quad e_{0,1}, \quad e_{1,2}, \quad r_0, \quad r_1$$

may be affected by their replacement, and the shared edge is center-free. Let $a, c, B, r \in [0, 1)$ be the boundary reach on $e_{5,0}$, the own-radial reach on r_0 , the boundary reach on $e_{1,2}$, and the radial demand on r_1 needed to overlap the unchanged center interval. If

$$a < \frac{1}{2}, \quad a + c < 1, \quad a + B < 1, \quad r < \max\{1 - a, 1 - B\},$$

there are two open unit equilateral replacements preserving these five skeleton pieces, with actual boundary sums below one and no positive adjacent support. No center-class formula is assumed.

Proposition D.16 (Vd1 input to the two-vertex replacement). *Assume a skeleton cover has a unique Vd1 role T_0 supplying M_1 , a unique supercritical role T_1 , and all other V roles are nonsupercritical Vd0. Suppose $e_{0,1}$ is center-free and no other V role has positive trace on r_0 or r_1 . The scalar replacement applies in CE1 and CE2 and yields a skeleton cover with $N'_+ = 0$.*

APPENDIX E. ZERO-GAP NINE-POINT OPTIMIZATION

This appendix proves the frontier construction, witness exclusions, and four-contact enclosure bound for the zero-gap obstruction. It also introduces inner comparison points for the exact certificate, without changing the three frontier witnesses Q_-, Q_0, Q_+ .

E.1. Corner-clipped AB -unions. In this subsection the corner is the local origin O_{loc} , not the hexagon center O . The local anchors A, B used here are also distinct from the Newton comparison points introduced later. The corner chart

$$\Psi_4(u, v) = V_4 + u(V_5 - V_4) + v(V_3 - V_4)$$

has coordinate axes corresponding to the forward and backward edge directions. We identify a local vector $ue_A + ve_B$ with its pair (u, v) when applying this chart.

Let e_A, e_B be unit vectors with angle 120° , put

$$O_{\text{loc}} = 0, \quad A = ae_A, \quad B = be_B, \quad C_{\text{loc}} = \{ue_A + ve_B : u, v \geq 0\},$$

and define

$$(51) \quad \mathcal{R}_{AB}^{\text{loc}}(a, b) = C_{\text{loc}} \cap \bigcup \left\{ T : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ O_{\text{loc}}, A, B \in T \end{array} \right\}.$$

By the wedge reduction, the global union from the body is $\Psi_4(\mathcal{R}_{AB}^{\text{loc}}(b, a))$; its two anchor parameters are exchanged.

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

Theorem E.1 (Strict supercritical AB -union). *Assume $a, b > 0$ and*

$$a + b > 1, \quad \rho = a^2 + ab + b^2 < 1,$$

put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, and define

$$\begin{aligned} \alpha &= \frac{h(a + 2b - aD)}{2\rho}, & \beta &= \frac{h(a - b + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-a + b + (a + b)D)}{2\rho}, & \delta &= \frac{h(2a + b - bD)}{2\rho}. \end{aligned}$$

Then all four coefficients are positive and

$$(52) \quad \mathcal{R}_{AB}^{\text{loc}}(a, b) = (\mathcal{C}_{\text{loc}} \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (\mathcal{C}_{\text{loc}} \cap \mathcal{D}_B \cap \mathcal{H}_B),$$

where

$$\begin{aligned} \mathcal{D}_A &= \{(u, v) : (u - a)^2 + v^2 - (u - a)v \leq 1\}, \\ \mathcal{D}_B &= \{(u, v) : u^2 + (v - b)^2 - u(v - b) \leq 1\}, \\ \mathcal{H}_A &= \{(u, v) : \alpha(u - a) + \beta v \leq 0\}, \\ \mathcal{H}_B &= \{(u, v) : \gamma u + \delta(v - b) \leq 0\}. \end{aligned}$$

Its non-axis frontier consists, in counterclockwise order, of the A -circle, the A -line, the B -line, and the B -circle.

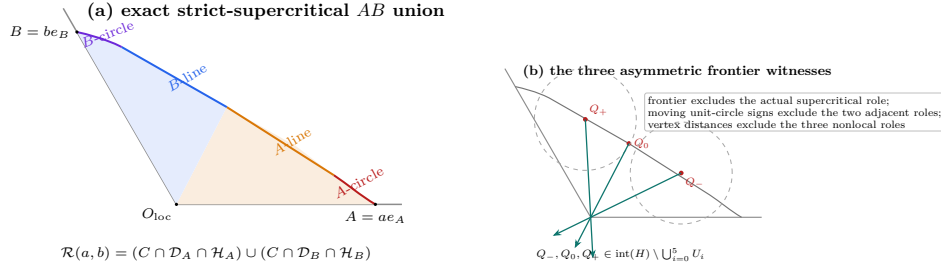


FIGURE 8. The strict-supercritical AB frontier and the three asymmetric witnesses. The source-conditioned union has, in counterclockwise order, an A -circle arc, an A -line segment, a B -line segment, and a B -circle arc. The points Q_-, Q_0, Q_+ are selected on this exact frontier and are then excluded from the six actual V roles by frontier, moving-circle, and vertex-distance arguments. The drawing is schematic; the labelled frontier order and witness roles are exact.

E.2. Exact-one handoffs and common reach bounds.

Lemma E.2 (Exact-one handoff chain). *Assume the six V triangles cover ∂H and exactly one actual V triangle is supercritical. Rotate its index to 4. The strict handoffs supplied by Proposition 2.12 satisfy*

$$\xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

With

$$a = 1 - \xi_3, \quad b = \xi_4, \quad p = 1 - b, \quad q = 1 - a,$$

one has

$$(53) \quad 0 < a, b < 1, \quad a + b > 1, \quad a^2 + ab + b^2 < 1,$$

and every selected V triangle obeys

$$a_i \geq p, \quad b_i \geq q.$$

E.3. Six directly forced radial points. Put $h = \sqrt{3}/2$ and

$$s = p + q, \quad m = \min(p, q), \quad M = \max(p, q), \quad \chi = s^4 - s^2 + pq.$$

The strict domain (53) gives

$$(54) \quad pq > (1 - s)(2 - s), \quad m > 1 - s, \quad s > 1 - m.$$

Let $c_* = c_{\max}(p, q)$ be the exact radial envelope of Proposition A.11. On the present subcritical domain, equation (29) becomes

$$(55) \quad c_* = \begin{cases} C_L(m), & \chi \leq 0, \\ \frac{2M}{1 + \sqrt{4s^2 - 3}}, & \chi > 0, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal s .

Lemma E.3 (Direct radial forcing). *One has*

$$(56) \quad 0 < c_* < 1, \quad c_* > 1 - m.$$

Assume that $U_C, U_0, \dots, U_5$ cover H , with no restriction on the normalized V types. Then, for every i ,

$$D_i = (1 - c_*)V_i \in U_C.$$

Consequently U_C contains the centered closed disk

$$(57) \quad \mathcal{D}_\eta = \{x : \|x\| \leq \eta\}, \quad \eta = h(1 - c_*).$$

We next define the three asymmetric witnesses. At V_4 use local coordinates

$$P = \Psi_4(u, v), \quad \mathbf{q}(u, v) = u^2 + v^2 - uv.$$

The forward-reach lower bound is b and the backward-reach lower bound is a . Accordingly, set

$$\begin{aligned} \alpha &= \frac{h(b + 2a - bD)}{2\rho}, & \beta &= \frac{h(b - a + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-b + a + (a + b)D)}{2\rho}, & \delta &= \frac{h(2b + a - aD)}{2\rho}, \end{aligned}$$

where now $\rho = a^2 + ab + b^2$ and $D = \sqrt{4\rho - 3}$. The frontier coefficients here are not the signed-center slacks. Applying Theorem E.1 with the exchanged pair (b, a) , the two local line pieces are

$$L_-(\lambda) = (b - \beta\lambda, \alpha\lambda), \quad L_+(\mu) = (\delta\mu, a - \gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D + 3))$ in

$$(58) \quad J = \left(\frac{2b + a - aD}{D + 3}, \frac{b + 2a - bD}{D + 3} \right).$$

Put

$$E_- = (1 - b, 2 - b), \quad E_+ = (2 - a, 1 - a).$$

The actual adjacent handoffs have local centers

$$Z_5(t) = (1 + t, t), \quad 1 - a \leq t \leq b, \quad \text{and} \quad Z_3(y) = (y, 1 + y), \quad 1 - b \leq y \leq a.$$

The restrictions of the two unit-circle equations are

$$(59) \quad \begin{aligned} g_-(\lambda) &= \frac{3}{4}\lambda^2 - 3(\alpha + b\beta)\lambda + 3b^2 - 3b + 2, \\ g_+(\mu) &= \frac{3}{4}\mu^2 - 3(\delta + a\gamma)\mu + 3a^2 - 3a + 2. \end{aligned}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\begin{aligned} \lambda_- &= 2(\alpha + b\beta) - \frac{2}{3}\sqrt{9(\alpha + b\beta)^2 - 9b^2 + 9b - 6}, \\ \mu_+ &= 2(\delta + a\gamma) - \frac{2}{3}\sqrt{9(\delta + a\gamma)^2 - 9a^2 + 9a - 6}. \end{aligned}$$

Lemma E.4 (Moving adjacent-triangle signs). *Throughout (53),*

$$(60) \quad \lambda_* < \lambda_- < h^{-1}, \quad \mu_* < \mu_+ < h^{-1}.$$

Moreover $J_u, J_v < 1/2$. The point $L_+(\mu_+)$ satisfies

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3 - 2a,$$

and the reflected bound holds for $L_-(\lambda_-)$. Writing $F(P, t) = \mathbf{q}(P - Z_5(t)) - 1$, one has, for every $t \in [1 - a, b]$,

$$(61) \quad F(L_+(\mu_+), t) \geq 0, \quad F(J, t) > 0, \quad F(L_-(\lambda_-), t) > 0.$$

Under the reflection $(u, v, a, b) \leftrightarrow (v, u, b, a)$, the analogous signs hold for every $y \in [1 - b, a]$: the first-root point $L_-(\lambda_-)$ has nonnegative sign against $Z_3(y)$, and the other two witnesses have strictly positive sign.

Define the local and global witnesses by

$$(62) \quad Q_-^{\text{loc}} = L_-(\lambda_-), \quad Q_0^{\text{loc}} = J, \quad Q_+^{\text{loc}} = L_+(\mu_+),$$

and put $Q_j = \Psi_4(Q_j^{\text{loc}})$ for $j \in \{-, 0, +\}$.

Lemma E.5 (Direct asymmetric forcing). *Assume that $U_C, U_0, \dots, U_5$ cover H . Then*

$$Q_-, Q_0, Q_+ \in \text{int}(H) \setminus \bigcup_{i=0}^5 U_i,$$

and consequently all three witnesses belong to U_C .

Together with lemma E.3, this proves the containment $K_{\text{wit}}(a, b) \subset U_C$ used in the body.

E.4. Newton inner reduction. For the nontrivial regime $c_* > 2/3$, we first remove the square radicals in the first-root witnesses. Normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D+3)}.$$

Define $\tilde{g}_-(x) = g_-(hx)$ and $\tilde{g}_+(x) = g_+(hx)$ from (59). Take one Newton step at the junction:

$$\hat{\xi}_- = \xi_* - \frac{\tilde{g}_-(\xi_*)}{\tilde{g}'_-(\xi_*)}, \quad \hat{\zeta}_+ = \zeta_* - \frac{\tilde{g}_+(\zeta_*)}{\tilde{g}'_+(\zeta_*)}.$$

Define the global Newton vectors by

$$\begin{aligned} A &= \Psi_4 \left(b - \frac{3}{4}\bar{\beta}\hat{\xi}_-, \frac{3}{4}\bar{\alpha}\hat{\xi}_- \right), \\ B &= \Psi_4(J), \\ C &= \Psi_4 \left(\frac{3}{4}\bar{\delta}\hat{\zeta}_+, a - \frac{3}{4}\bar{\gamma}\hat{\zeta}_+ \right), \end{aligned}$$

All norms and cross products below are taken in these global vectors.

Lemma E.6 (Newton inner reduction). *Their local coordinates are rational in a, b, D , and their global coordinates are rational over $\mathbb{Q}(\sqrt{3})$ in those variables. They satisfy*

$$A \in (Q_0, Q_-), \quad C \in (Q_0, Q_+).$$

Consequently

$$\hat{K} = \text{conv}(\mathcal{D}_\eta \cup \{A, B, C\}),$$

satisfies

$$(63) \quad \hat{K} \subseteq \text{conv}(K_{\text{wit}}(a, b)), \quad \Lambda(K_{\text{wit}}(a, b)) \geq \Lambda(\hat{K}).$$

E.5. Ordered convex chain and supporting-line bounds.

Proposition E.7 (Four-contact witness geometry). *Assume (53) and $c_* > 2/3$. The rays A, B, C occur in that order in one open 120° cone, and*

$$\text{cross}(B - A, C - B) > 0, \quad \|A\|, \|C\| > \eta.$$

Moreover the outward distances of the lines AB, BC satisfy

$$(64) \quad d_{AB} := \frac{\text{cross}(A, B)}{\|B - A\|} \geq h - 2\eta, \quad d_{BC} := \frac{\text{cross}(B, C)}{\|C - B\|} \geq h - 2\eta.$$

E.6. Four supporting contacts and paired radius transfer. Let R and J denote counterclockwise rotations through $2\pi/3$ and $\pi/2$, respectively. For the inner witness define

$$H(n) = \max\{\eta, \langle A, n \rangle, \langle B, n \rangle, \langle C, n \rangle\}, \quad \Psi(n) = \sum_{j=0}^2 H(R^j n).$$

The support formula gives $\Lambda(\hat{K}) = h^{-1} \min_{\|n\|=1} \Psi(n)$. The minimum can be located without introducing another convex body.

Lemma E.8 (Four-contact enclosure formula). *For the Newton inner witness in the range $c_* > 2/3$, put*

$$\begin{aligned} n_{AB} &= -\frac{J(B-A)}{\|B-A\|}, & n_{BC} &= -\frac{J(C-B)}{\|C-B\|}, \\ t_A &= \frac{\eta A - \sqrt{\|A\|^2 - \eta^2} JA}{\|A\|^2}, & t_C &= \frac{\eta C + \sqrt{\|C\|^2 - \eta^2} JC}{\|C\|^2}. \end{aligned}$$

Then

$$(65) \quad \Lambda(\hat{K}) = \frac{1}{h} \min\{\Psi(n_{AB}), \Psi(n_{BC}), \Psi(t_A), \Psi(t_C)\}.$$

At the first two contacts the disk supplies the other two supports, so

$$(66) \quad \Psi(n_{AB}) \geq d_{AB} + 2\eta \geq h, \quad \Psi(n_{BC}) \geq d_{BC} + 2\eta \geq h.$$

For the two tangencies, put

$$\Delta = \text{cross}(C, RA) > 0, \quad \nu = \langle C, RA \rangle, \quad N_X = \|X\|^2 \quad (X = A, C).$$

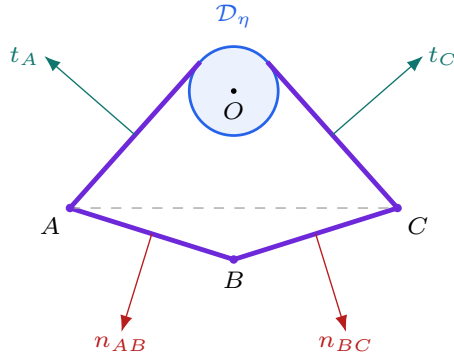
Choosing C as a support source at Rt_A and A at R^2t_C , with the disk in the other directions, gives

$$(67) \quad \Psi(t_A) \geq 2\eta + \frac{\eta\nu + \Delta\sqrt{N_A - \eta^2}}{N_A}, \quad \Psi(t_C) \geq 2\eta + \frac{\eta\nu + \Delta\sqrt{N_C - \eta^2}}{N_C}.$$

Consequently the only remaining targets are the paired residual signs

$$(68) \quad P_X(\eta) := (N_X - \eta^2)\Delta^2 - ((h - 2\eta)N_X - \eta\nu)^2 \geq 0 \quad (X = A, C).$$

(a) the four exposed contacts



(b) four support sums

Two consecutive-point contacts:

$$\Psi(n_{AB}) \geq d_{AB} + 2\eta \geq h,$$

$$\Psi(n_{BC}) \geq d_{BC} + 2\eta \geq h.$$

Two exposed disk tangencies:

$$P_A(\eta), P_C(\eta) \geq 0$$

$$\implies \Psi(t_A), \Psi(t_C) \geq h.$$

$$\text{Hence } \Lambda(\hat{K}) \geq 1.$$

FIGURE 9. The four-contact terminal for the zero-gap witness. The supporting lines through AB and BC and the two outer disk tangencies exhaust the active changes of support source. The dashed segment AC is not exposed: B and the disk center lie on opposite sides. The drawing is schematic; the exact witness geometry and the paired tangent residuals are proved in the text. No auxiliary Minkowski sum or cap covering is needed.

Indeed $P_X(\eta) \geq 0$ gives $\Delta\sqrt{N_X - \eta^2} \geq |(h - 2\eta)N_X - \eta\nu|$, which implies the required lower bound even when the expression inside the absolute value is negative. No sign is lost by squaring.

The exact certificate uses a smaller rationally specified radius. The following elementary transfer is the reason those existing signs suffice for the actual-radius tangencies.

Lemma E.9 (Simultaneous radius transfer). *Let X, Y be vectors, $d \geq 0$, and $0 \leq e_0 \leq 1/3$. If*

$$\|(1 - 2e_0)X - e_0Y\| \leq d, \quad \|(1 - 2e_0)Y - e_0X\| \leq d,$$

then both inequalities hold at every $e \in [e_0, 1/3]$.

Lemma E.10 (Solved witness-enclosure optimization). *For every parameter pair in equation (16),*

$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

APPENDIX F. EXACT DISK-TANGENT RESIDUAL CERTIFICATE

The two disk-tangent contact bounds in the zero-gap nine-point theorem are certified by exact symbolic arithmetic. This section gives the mathematical reduction and verification algorithms. The finite sparse data and the two exact programs form the electronic appendix at an immutable revision; they are included in the accompanying source package. The PDF prints the certificate construction and checking rules, not every machine coefficient. No floating-point arithmetic, interval arithmetic, adaptive subdivision, branch-and-bound, or unrecorded numerical tolerance is used.

F.1. Certificate manifest. The machine-readable provenance manifest identifies the six sparse-data shards and both exact verifiers by full Git blob and records their roles. They decode to one canonical sparse-polynomial transcript with SHA-256 digest

dc46aaf263655d5159ecd3a81db72ee82477951d06172f4743b248df37209485

The derivation verifier reconstructs the transcript from the witness formulas and compares every coefficient with the authenticated object. The positivity verifier derives the Bernstein coefficients from that object and checks their signs exactly.

F.2. Rational radial upper envelopes. Retain the notation of Section E. Thus

$$p = 1 - b, \quad q = 1 - a, \quad s = p + q, \quad m = \min\{p, q\}, \quad M = \max\{p, q\},$$

$$\chi = s^4 - s^2 + mM, \quad h = \frac{\sqrt{3}}{2}, \quad c_* = c_{\max}(p, q).$$

The strict witness domain gives

$$(69) \quad mM > (1 - s)(2 - s), \quad m > 1 - s, \quad s > \frac{2}{3}.$$

Put

$$F_m(x) = x^4 - x^2 + mx - m^2, \quad \kappa = F'_m(s) = 4s^3 - 2s + m.$$

Then

$$\kappa > 4s^3 - 3s + 1 = (s + 1)(2s - 1)^2 > 0.$$

Define

$$(70) \quad \bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \leq 0), \quad \bar{c}_T = s - \frac{\chi}{1-m} \quad (\chi \geq 0).$$

Lemma F.1 (Branchwise radial upper bounds). *On the corresponding radial cells,*

$$c_* \leq \bar{c}_L < 1, \quad c_* \leq \bar{c}_T < 1.$$

At $\chi = 0$ both upper bounds equal $s = c_*$.

Reflect so that $z = a - b \geq 0$ and put $U = a + b - 1$. Then

$$s = 1 - U, \quad m = \frac{1 - U - z}{2}, \quad D^2 = z^2 + 6U + 3U^2 =: K(U, z),$$

and

$$\begin{aligned} \Xi(U, z) &= 4\chi = 1 - 10U + 21U^2 - 16U^3 + 4U^4 - z^2, \\ G(U, z) &= 2\kappa = 5 - 21U + 24U^2 - 8U^3 - z. \end{aligned}$$

Thus

$$(71) \quad \bar{c}_L = 1 - U - \frac{\Xi(U, z)}{2G(U, z)}, \quad \bar{c}_T = 1 - U - \frac{\Xi(U, z)}{2(1 + U + z)}.$$

F.3. Gram reduction and the geometric implication. On the active cell let $\bar{c}$ denote the corresponding expression in (71), and put

$$\bar{\eta} = h(1 - \bar{c}), \quad e = 1 - \bar{c}, \quad t = 2\bar{c} - 1.$$

Lemma F.1 gives $0 < \bar{\eta} \leq \eta = h(1 - c_*)$. The algebra below is evaluated at $\bar{\eta}$; lemma E.9 will transfer both residual signs to the actual radius η .

Represent a global vector as (x, hy) and put $C' = R^{-1}C$. Define

$$N_A = \|A\|^2, \quad N_C = \|C\|^2, \quad \nu = \langle A, C' \rangle,$$

$$\Delta_0 = x_{C'}y_A - y_{C'}x_A > 0.$$

Thus $h\Delta_0 = \text{cross}(C', A) = \text{cross}(C, RA) > 0$. The authenticated code uses the opposite signed determinant, but every occurrence is squared; its integer polynomials and digest are unchanged. The Gram identity is

$$(72) \quad h^2\Delta_0^2 = N_A N_C - \nu^2.$$

Define

$$(73) \quad Q_A = \Delta_0^2 - \|tA - eC'\|^2, \quad Q_C = \Delta_0^2 - \|tC' - eA\|^2.$$

Lemma F.2 (Residual signs imply the actual-radius tangent bounds). *If $Q_A, Q_C \geq 0$, then the two actual-radius residuals $P_A(\eta), P_C(\eta)$ in equation (68) are nonnegative. Consequently both tangent contact sums in equation (67) are at least h .*

F.4. Eight exact integer-polynomial signs. The Newton witnesses A, C are rational functions of U, z, D . Substituting (71) into (73), clearing denominators, and reducing by

$$D^2 - K(U, z) = 0$$

gives, for $B \in \{L, T\}$ and $X \in \{A, C\}$,

$$(74) \quad \text{num}(Q_{B,X}) = 32(K+3)^2 \{A_{B,X}(U, z) + DB_{B,X}(U, z)\},$$

where all eight core polynomials lie in $\mathbb{Z}[U, z]$. Every omitted denominator is strictly positive on the strict domain: its factors are positive constants, $(D+3)^4$, $G(U, z)^2$ or $(1+U+z)^2$, and squares of the two Newton-derivative numerators. Here $G = 2\kappa > 0$, and the fixed-line sign theorem proves both Newton derivatives strictly negative.

The derivation verifier constructs A, C , both rational envelopes, and all four residuals directly in the exact field $\mathbb{Q}(U, z, D)$. It performs the reduction modulo $D^2 - K$, removes the common positive factor, primitive-normalizes the eight integer polynomials, and compares their sparse term lists coefficient by coefficient with the authenticated transcript. Thus the data are checked against the geometric formulas rather than trusted as an independent precomputation.

F.5. Three global positive-basis certificates. For integers $0 \leq i \leq m$, the Bernstein basis polynomial is $B_i^m(r) = \binom{m}{i} r^i (1-r)^{m-i}$. A tensor Bernstein expansion on $[0, 1]^2$ has the form

$$P(r, s) = \sum_{i=0}^m \sum_{j=0}^n b_{ij} B_i^m(r) B_j^n(s).$$

Every basis term is nonnegative, and their sum is one. Therefore nonnegative coefficients certify $P \geq 0$ on the whole square; strictly positive coefficients certify strict positivity. The exact coefficient conversion formula is given below.

Set

$$\mathcal{G}(U) = 1 - 6U - 3U^2, \quad \mathcal{P}(U) = 1 - 10U + 21U^2 - 16U^3 + 4U^4,$$

$$U_{\Delta} = 1 - \frac{\sqrt{3}}{2}, \quad U_{\max} = \frac{2}{\sqrt{3}} - 1.$$

The complete reflected domain is the union of the three closed cells

cell	U -range	z^2 -range
T	$0 \leq U \leq U_{\Delta}$	$0 \leq z^2 \leq \mathcal{P}(U)$
L_0	$0 \leq U \leq U_{\Delta}$	$\mathcal{P}(U) \leq z^2 \leq \mathcal{G}(U)$
L_1	$U_{\Delta} \leq U \leq U_{\max}$	$0 \leq z^2 \leq \mathcal{G}(U)$

For the T cell use

$$U = \frac{(x-1)(3-x)}{4x}, \quad z = y \frac{9-x^4}{8x^2}, \quad 1 \leq x \leq \sqrt{3}, \quad 0 \leq y \leq 1,$$

and then $x = 1 + (\sqrt{3} - 1)r$. This maps $[0, 1]^2$ onto the complete cell. After multiplication by positive powers of 2 and x , the four transformed core polynomials

have global tensor Bernstein expansions with data

	bidegree	coefficient count	zero count
$A_{T,A}$	$(88, 22)$	2047	2
$B_{T,A}$	$(80, 20)$	1701	0
$A_{T,C}$	$(88, 22)$	2047	2
$B_{T,C}$	$(80, 20)$	1701	0.

Every nonzero coefficient is strictly positive.

For an L -cell polynomial write

$$F(U, z) = E_F(U, w) + zO_F(U, w), \quad w = z^2,$$

and

$$R_F(U, w) = E_F(U, w)^2 - wO_F(U, w)^2.$$

The implications $E_F \geq 0$ and $R_F \geq 0$ give $E_F \geq \sqrt{w}|O_F|$, hence $F \geq 0$ for $z \geq 0$. Use the exact charts

$$U = U_\Delta r, \quad w = \mathcal{P}(U) + s\{\mathcal{G}(U) - \mathcal{P}(U)\}$$

for L_0 , and

$$U = U_\Delta + (U_{\max} - U_\Delta)r, \quad w = s\mathcal{G}(U)$$

for L_1 . The positivity verifier forms one global Bernstein expansion for each of E_F, R_F on each square. The authenticated transcript records the bidegrees and coefficient counts of all sixteen expansions; every coefficient is strictly positive.

The verifier implements polynomial addition, multiplication, substitution, and power-to-Bernstein conversion over exact integers. Coefficients in $\mathbb{Q}(\sqrt{3})$ are represented as rational pairs and signed by exact rational comparisons. For $x + y\sqrt{3}$ with rational x, y , equal signs are immediate. When $x > 0 > y$, its sign is that of $x^2 - 3y^2$; when $y > 0 > x$, it is that of $3y^2 - x^2$. Zero components are handled directly. Thus no numerical approximation to $\sqrt{3}$ is needed. If $P(r, s) = \sum a_{ij}r^i s^j$ has bidegree (m, n) , its exact tensor Bernstein coefficients are

$$b_{k\ell} = \sum_{i \leq k} \sum_{j \leq \ell} a_{ij} \frac{\binom{k}{i}}{\binom{m}{i}} \frac{\binom{\ell}{j}}{\binom{n}{j}}.$$

This is precisely the conversion implemented by the verifier.

Theorem F.3 (Exact tangent-residual certificate). *On the complete strict witness domain with $c_* > 2/3$, all eight polynomials in (74) are nonnegative. Hence $Q_A, Q_C \geq 0$, and the two actual-radius tangent contact bounds in equation (67) hold.*

Remark F.4 (Reproduction). The authenticated data and the two verifiers constitute the exact electronic certificate used by the proof. Reproduce it from its package directory by executing

```
python verify_mixed_overlap_core_derivation.py
python verify_global_core_positivity.py
```

Both commands use exact arithmetic and must reproduce the transcript digest above and the asserted sign checks.